

Solving a new composition from a converged wind, and what the base handoff carries

1. What this memo records

Two things that are easy to confuse with each other.

Part 1: the composition restart. How a converged EXHALE solution is used as the starting guess for a *different* composition: what the run script does, what load_IC does to the loaded densities, where the seed's information is kept and where it is overwritten, and (the point that matters for how results are quoted) why the seed sets only the path and never the answer. Everything in this part is implemented, verified against the source, and in production use in the LHS 1140 b composition and K_{zz} scans.

Part 2: the base handoff. EXHALE now has two handoff routes: the scalar `base.inp` and the profile file named by `Lower atmosphere profile:.` This part records what each one is in the code today, the comparison on which the choice between them was made, and (item by item against the plan's requirement list) what the profile route actually carries, what reads each field, and what is still not transmitted.

Decision (2026-08-26), implemented 2026-08-27. Phase E goes to a profile file. The scalar `base.inp` is not retired: it stays as the single-layer route, and its behavior with no profile beside it is unchanged. What the decision left to the implementation (the schema, and what happens when both files are present) is now written, and a scalar physics key beside a profile is refused rather than merged (`src/modules/files_IO/input_read.f90:1371-1386`).

Part 1: solving a new composition from a converged wind

2. The recipe, as the run script executes it

The production example is `LHS1140b/exhale/run_heh_diff_case.sh`, which solves one point of the helium-abundance scan with the binary H/He diffusion operator active. In order (line numbers are that file):

1. **Clone the input, change one line.** The control case's `input.inp` is copied with `sed` and exactly the composition line is replaced (lines 27–31): `He/H number ratio:` takes the new value, while the planet tag, the direct-steady switch and the residual target are set for this run and `He_Kzz:` is appended (line 32). Nothing else about the configuration moves (same spectrum, same domain, same flags), so the scan is a scan in composition and not in anything else.
2. **Hand the converged seed over as the initial condition** (lines 33–34): the seed run's `output/Hydro_ioniz.txt` and `output/Ion_species.txt` are copied into the new run directory as `output/Hydro_ioniz_IC.txt` and

output/Ion_species_IC.txt, which are the two file names load_IC reads (src/modules/files_IO/load_IC.f90:2-4, 89, 106, 126).

3. **Solve the steady state directly.** The binary is run with EXHALE_PTC=1, EXHALE_PTC_JFNK=1 and EXHALE_PTC_DTAU0=1.0 (line 43), i.e. the JFNK direct-steady route rather than a marching relaxation.
4. **Feed the output back as the next initial condition**, when the diffusion operator is on (lines 40–41): each further invocation copies the run's own last output over its *_IC.txt files. This is the outer Picard loop of composition and wind driven from the outside, one pass-set at a time, and the loop is stopped when the reported composition drift stops falling.
5. **Post-processing and transit** (lines 51–57): one Do only PP: True pass to write the advection-corrected profiles, then EXHALE_transit.py for the He I 10830 line.

Inside a single invocation the same Picard loop runs up to twenty times when He_diffusion is on and exactly once when it is off (src/EXHALE_main.f90:1238); it exits on a composition drift below 10^{-3} (line 1268) or on a failed steady solve (line 1254), and the composition update is under-relaxed with ω starting at 0.5 and halved down to a floor of 0.125 on any pass whose drift failed to fall (lines 1229, 1270–1274). The drift tested is the undamped distance to the fixed point, so a small ω cannot buy a false convergence (lines 1213–1214).

3. What the restart does to the composition

The seed file holds *absolute* species densities. Read as they stand, they would silently run the donor's composition while the setup report echoed the input one. The input file is the authority, so load_IC carries the loaded hydrogen and helium onto the requested ratio (src/modules/files_IO/load_IC.f90:428–443).

3.1 Counting the elements

In each cell the two element densities are counted over every species that carries a nucleus, the molecular carriers and the oxygen-chemistry species included (load_IC.f90:444–452):

$$n_{\text{H}} = n(\text{H I}) + n(\text{H II}) + 2[n(\text{H}_2) + n(\text{H}_2^+)] + 3n(\text{H}_3^+) + n(\text{HeH}^+) + n(\text{OH}) + 2n(\text{H}_2\text{O}), \quad (1)$$

$$n_{\text{He}} = n(\text{He I}) + n(\text{He II}) + n(\text{He III}) + n(\text{HeH}^+). \quad (2)$$

HeH⁺ is inside both counts, because it carries a nucleus of each. The metastable 2³S state is *not* a term of its own: the He I column the code transports and the file carries is the total He I population, the triplet included, so its nucleus is already counted there and adding it again would count it twice (load_IC.f90:449–450).

These are the integer weights bsp_nH and bsp_nHe of the species table (src/modules/init/species_table.f90: 155–160), over the thirteen base species HI HII HeI HeII HeIII HeTR H2 H2+ H3+ HeH+ OH H2O CO:

	HI	HII	HeI	HeII	HeIII	HeTR	H ₂	H ₂ ⁺	H ₃ ⁺	HeH ⁺	OH	H ₂ O	CO
bsp_nH	1	1	0	0	0	0	2	2	3	1	1	2	0
bsp_nHe	0	0	1	1	1	1	0	0	0	1	0	0	0

taken together with `bsp_is_excited_level, .true.` for HeTR alone, which is what every budget sum skips (`species_table.f90:185-189`). The same count is what `element_ratio_HeH` applies in the code proper (`src/modules/functions/composition.f90:374-402`) and what the diffusion operator writes back against (`element_nucleus_counts`, called from `mixture_mass_split`, `src/modules/functions/binary_element_diffusion.f90:3041-3106`). One element count, one definition, three call sites: there is no second formula that can drift away from this one. That the triplet had to be excluded was measured, not assumed: it was being added a second time in all three places and in four more, and the mass, particle, nucleus and collision-partner sums were each wrong by that amount until 2026-08-27 (`docs/Update_EXHALE_stage1.pdf §75`).

3.2 The redistribution

Let `HeH` be the requested ratio. The deviation of the loaded state is measured cell by cell as $|n_{\text{He}}/n_{\text{H}} - \text{HeH}|/\text{HeH}$ and the redistribution is triggered when its maximum exceeds `heh_dev_tol = 10-6` (`load_IC.f90:479-493`). A restart at the composition the file was written with therefore leaves every density untouched (bit for bit), which is what keeps the same-composition restart path exact.

The comparison runs over the physical cells alone, and it names the cell it attained. Two properties of that loop were measured into it and both matter to anyone reading a refusal message. First, the range is $j = 1 \dots N$: the ghosts are boundary data, the lower pair being the inflow reservoir whose molecular partition the handoff imposes and the sweep re-pins every step, and the upper pair mirroring the top cell, so a check on the input's He/H has nothing to say about them. The reason it is not the whole array is a measured incident (2026-09-05): a carrier-transport snapshot carried its lower ghost with He/H 2.1×10^{-5} off the input while every physical cell agreed to 10^{-6} , and that ghost alone sent the restart into the rescale branch, which the `HeH+` clause below then refused. Second, the quantity that decides is the *largest* departure over those cells, and `report_deviating_heh_cell` prints the cell that attained it, with its radius, the He/H the file carries there, the He/H the input asks for, the relative departure and the threshold, all at ES16.9. The message used to print the top physical cell instead, which on a molecular state is not the deciding cell at all: MEASURED on a coupled-carrier state, the top cell agreed with the input to seventeen digits while cell 199 at $r = 1.0906$ was off by 3.96×10^{-2} relative, so the refusal read “7.9307E-02 against 7.9307E-02 ... cannot be rescaled”. Two numbers that print the same cannot explain a refusal.

When it does trigger, the two scale factors are (`load_IC.f90:514-515`)

$$s_{\text{H}} = \frac{n_{\text{H}} + n_{\text{He}}}{(1 + \text{HeH})n_{\text{H}}}, \quad s_{\text{He}} = \text{HeH} \frac{n_{\text{H}} + n_{\text{He}}}{(1 + \text{HeH})n_{\text{He}}}, \quad (3)$$

so that the new element densities in that cell are

$$n'_{\text{H}} = \frac{n_{\text{H}} + n_{\text{He}}}{1 + \text{HeH}}, \quad n'_{\text{He}} = \text{HeH} \frac{n_{\text{H}} + n_{\text{He}}}{1 + \text{HeH}}, \quad n'_{\text{H}} + n'_{\text{He}} = n_{\text{H}} + n_{\text{He}}. \quad (4)$$

That is the defining property of the operation: **the total nuclei count of each cell is conserved** and only the split between the two elements moves. Every hydrogen-bearing column is multiplied by s_{H} and every helium-bearing one by s_{He} (`load_IC.f90:530-547`), the trace metals following hydrogen because they are defined relative to a hydrogen nucleus, and OH, H₂O and CO following it too because among the two elements being rescaled they carry only H (their oxygen and carbon follow hydrogen through `melem_ab`, like every trace metal), so

- the **ionization-stage split inside each element** is preserved exactly: $n(\text{H II})/n_{\text{H}}$, $n(\text{He II})/n_{\text{He}}$ and the 2^3S fraction all come through unchanged, since a single factor multiplies every stage of the element;
- the **velocity, pressure and temperature** profiles are the seed's, read straight off `Hydro_ioniz_IC.txt`;
- the **mass density is not** read from the file. It is rebuilt by `calc_rho` from the redistributed densities under the run's own mass policy, $\text{He } 2^3S$, molecular and (under `eos_metals`) metal mass included (`load_IC.f90:778-800`). It therefore does move with the composition, as it must: swapping hydrogen for helium at fixed nuclei count changes the mass. The seed supplies the kinematics and the thermal state; the mass follows the new composition.

3.3 Two refusals, and the diffusion-on exception

HeH^+ carries one nucleus of each element, so no single factor can set both counts. Two cases are therefore refused outright rather than approximated (`load_IC.f90:494-513`): a file carrying no helium at all (nothing to rescale), and a molecular file carrying HeH^+ whose composition disagrees with the input while `He_diffusion` is off: such a state must be restarted at its own composition.

With `He_diffusion` on the rule changes, because the cell-by-cell element split *is* the state being restarted: the diffusion operator produced it, and a separated profile is the physics, not a defect of the file. The rescale is then confined to the base cells $1-N_g \dots 1$, which is exactly where the operator itself holds a Dirichlet reservoir composition; every cell above keeps the helium fraction it was written with, $s_{\text{H}} = s_{\text{He}} = 1$ (`load_IC.f90:516-529`). In those base cells HeH^+ is projected onto the two element totals the same way `binary_element_diffusion` writes back: it takes the smaller of the two factors, $\min(s_{\text{H}}, s_{\text{He}})$, and whatever nuclei that leaves short of n'_{H} and n'_{He} are deposited into the neutral ground species `H I` and `He I` (`load_IC.f90:548-576`; the same projection at `project_elements`, `binary_element_diffusion.f90:2107-2266`). The projection is non-negative by construction (the smaller factor can only reduce HeH^+), and it conserves both element totals, at the cost of moving a little of the molecular ion into the neutral species.

Either way the run says on `stdout` which of the two it did: with `He_diffusion` on it prints the kept top-cell ratio and the input ratio, and otherwise it prints the input ratio followed by the deviating cell's line (`load_IC.f90:574-585`), so the restart is never silent.

3.4 Elements the file does not carry

Separately from the composition, an element whose stages the file does not carry (no column at all, or columns identically zero because the run that wrote them had metals off) is built from its abundance exactly as a cold start does, $n_X = \text{ab}_X n_{\text{H}}$ with everything in the neutral stage, and the substitution is reported (`load_IC.f90:589-648`). The column test alone would accept a metals-off file and then run with zero metal density while the base boundary condition still counted metals in the mass and particle budget, leaving an $O(1)$ residual in the first cells; the zero-sum test is what closes that.

3.5 Elements whose reservoir the handoff has moved

A third case sits between the two above and was added on 2026-08-27: the file carries the element, but the *reservoir* has moved since the file was written. That is exactly what a flux-closure iteration does (it changes the elemental ratios of the lower atmosphere from one iteration to the next), so the base boundary condition would use the new El/H while every cell above the base still held the old one, and the elemental budget $n_{\text{El}}/n_{\text{H}} = (\text{El}/\text{H})$ could not close above the first cell (docs/Update_EXHALE_stage1.pdf §79–80).

The rule is a single factor on the whole column (load_IC.f90:650–705):

$$r_{\text{El}} = \frac{(\text{El}/\text{H})_{\text{handoff}}}{(\text{El}/\text{H})_{\text{loaded, base cell}}}, \quad (5)$$

applied to every ionization stage of that element alike, so the loaded ionization split and the radial shape of the profile cancel out of it and only the normalization moves. El/H at the base cell is counted in nuclei (the element summed over its stages against the hydrogen nuclei of equation (1)), which is the convention of `element_ratio_HeH` and of `src/utls/element_budget.py`, so the factor is measured in the units the budget check reads it back in.

Which elements are touched is not a test on the numbers. An element is rescaled if and only if the lower-atmosphere handoff *states* its reservoir, and what records that is the door the value came in through: `set_element_abundance` is the only routine either handoff reader uses to set an abundance, and passing through it sets `memem_from_handoff(e)` (`set_element_abundance` in `src/modules/files_IO/input_read.f90`). An abundance that reached the run from `metals.inp` alone never passes through that door, so it is not marked and its column is loaded untouched; a restart with no handoff at all leaves the loop doing nothing at all, which is what keeps every pre-existing restart bit for bit what it was. Three further guards skip the element: one already rebuilt from the abundance by the block above (it is the reservoir already), a zero abundance, and a loaded column that sums to zero at the base cell.

Helium is deliberately outside this rule and keeps the older convention of sections 3.: the base cells are set to the input He/H and, with `He_diffusion` on, the column above keeps its diffused split.

Each rescaled element prints its own line: loaded ratio, stated ratio, factor, at enough digits that a factor of 1.0001 is legible, because that is the size of the step a converged closure iteration takes (load_IC.f90:701–704).

4. The restart contract: what a state file states, and what is refused

Everything above is what the loader does to the numbers. This section is what the loader checks before it does any of it. A state is a solution of an equation set on a grid at a composition, so a file and a run that do not agree on those three things are not a restart pair at all, and since 2026-09-09 the disagreement is named rather than absorbed.

4.1 What a restart is for: Restart intent

A loaded state can be continued in three senses, and they are not variants of one path: a physical trajectory carries the clock the file records and every step obeys the physical-mode contract; a relaxation carries no clock and marches toward stationarity; a stationary evaluation takes no step at all and measures the state as it stands. One selector says which,

Restart intent: trajectory | relaxation | stationary [evaluate|equilibrate]

because a run that does not state its purpose cannot be checked against it (src/modules/files_IO/input_read.f90). The second word belongs to stationary alone: evaluate returns the state unchanged with its certification and takes no solver step, equilibrate puts the loaded composition on its own fixed point before the evaluation, which is a change of the state and is reported when it is taken, and the bare stationary measures the state as loaded and then enters the stationary solve.

Four consistency checks refuse rather than reinterpret, because a key that is silently ignored is a run doing something other than what its input file says: the key with Load IC? False (there is no loaded state for the intent to be about), Restart option change with the same (no state is being restarted), trajectory with Run mode: init (a trajectory is a physical integration and an initialization run claims no elapsed time), and stationary without Solver: Newton (there is no stationary solve to enter). The default is the run mode's own meaning: trajectory in phys, relaxation otherwise, which is what every restart did before the key existed, so no existing input changes behavior.

stationary_state_of_the_loaded_restart (src/EXHALE_main.f90) is the route that measures the state as loaded. Every earlier path into the solver went through the marching loop, which takes at least two CFL steps before it hands off, and a step is a different state, so what used to be certified was never the state the file carried. This route takes no step: it fixes WENO3, rebuilds the derived quantities in the checkpoint order (composition, then boundary, then primitive state), runs one equilibrium sweep because that sweep is part of the residual's definition, prints the loaded composition's departure from that sweep's root, and holds the residual and the certification to the file's own certified=field. MEASURED: the certified wasp_full_newton state is not the fixed point of its own sweep (the particle count moves 2.1×10^{-11} , the species 5.0×10^{-10} , the cooling 3.6×10^{-10} , which is the cell solve's $x_{\text{tol}} = \sqrt{\epsilon}$ and the file's digits), and that is reported at every evaluation rather than equilibrated away.

4.2 The metadata block, and the six fields it compares

Both state files carry an eight-line block of # comments, built and parsed by one routine each (build_restart_metadata and verify_restart_metadata, src/modules/files_IO/load_IC.f90), with every number written ES23.16 so that a field compared as text is compared as the number it stands for. Every reader of these files skips # lines, so the block changes no numeric parse and no regression verdict.

Field	What it states	On disagreement
restart_schema	the block's own version	refuse
reservoir	He/H and every trace El/H the run carries	refuse, naming the element
species_columns	the columns the state carries	report
grid	$N, R_0, r(1), r(N)$, the grid mode	refuse
constants	the constant set, R_J, k_B	refuse
options	20 physics switches, token by token	refuse, naming the token
t_phys[s]	the clock of the state	adopted if the coupling line lacks it
source	the build that produced it	report

The division is the point. The options field carries every switch that adds or removes an equation, an unknown or a source term, so a difference means the file's state solves another system; the numeric inputs (fluxes, tolerances, rates) are deliberately *not* there, because they change the coefficients of the same system and a state carried across such a change is a legitimate starting point whose residual simply is not zero. The species_columns line is reported and never refused, because species are restored by the label on the # columns line and an element the file does not carry is built from its abundance (§3.). A pair whose two halves carry different blocks is refused outright: they were not written by one run, so they are not one state.

A file with no block is loaded, and marked. A state written before the block existed is loaded exactly as it was, and the run says so: the configuration it was produced under cannot be compared, so every file that run writes carries restart_input=provenance_unknown in its source field. The mark is a property of the state and is inherited, because a state descended from an unknown configuration is no better known at the second reload than at the first, and dropping the mark there would lose it for good.

The grid belongs to the run. verify_restart_radii compares the cell centers of the physical cells $1 \dots N$ to 10^{-10} relative and refuses any other grid, naming the worst cell and both radii. The reason is that define_grid builds the faces, the widths and the window indices $j_{\min}$, $j_{\max}$ together with the centers, and a state file carries only the centers, so a file written under a different Grid type, Base grid or Outer radius passes a row-count guard whenever it has the same Grid cells and would then be marched on one grid's centers with another grid's faces. The tolerance is chosen against two measurements: the smallest real grid difference in the tree is 3.3×10^{-4} , and the coarsest writer of a state file, the Wind-AE initial condition at ES17.10, has a worst relative round-off of 5.0×10^{-11} , half of the tolerance. This reader does not interpolate.

4.3 Naming what a restart may change: Restart option change

The exact options comparison forbids the workflow this project actually uses to reach its solutions, which is to converge without an option and restart with it on. The key Restart option change: <token>[, <token> ...] names the tokens that are allowed to differ; everything else still refuses by name, with the token and its from -> to. Six layout tokens (metals, mol, oxychem, carrier, carrier_newton, iontrans) may never be named, because they decide which rows the state carries and naming one cannot make the file's rows this run's rows: that is a cold start, not a restart. An unknown token, a grid, reservoir or constant word, the key with no token, and the key without Load IC? True are all input errors.

A permitted change is written into both halves of the new state as # option_change from -> to at restart of <source> and inherited, up to a history of 32 lines, and EXHALE_setup.out gains a Restart provenance: line reading UNKNOWN, UNKNOWN by descent or states its configuration together with the changes this run permits.

What the state files no longer report. Two fields left the coupling header when the base boundary became the characteristic face condition: valve and fluxconst were the two switches the old ghost-cell boundary chose for itself, and the present boundary has neither, so a header reporting them would be reporting settings the run cannot have. The

loader's parser reads the fields it recognizes one at a time and ignores the rest, so a file written before that change still restores its `sec_ion` state and an older reader still finds `sec_ion` in a file written after it.

5. The seed is a guess; the answer is the new steady state

This is the distinction the whole recipe rests on, and it is worth stating plainly because the two are easy to conflate when a scan is quoted as a table of restarts.

The redistribution of section 3. is *arithmetic on a starting guess*. It is not a solution at the new composition, and it is not claimed to be one: it holds (v, p, T) and every ionization-stage ratio at the donor's values, which are the values that were in equilibrium with the *donor's* composition. What makes the result a result is what happens afterwards: the JFNK steady solve, the ionization-equilibrium solve, and (with diffusion on) the composition outer loop, iterated until the wind residual meets its target and the composition drift stops moving. The convergence *path* is what the seed choice buys; the converged state is a property of the new configuration alone.

The operational consequence: a restart chain is legitimate exactly as long as each rung is converged on its own terms, and each row of a scan must be quoted with its own achieved residual and drift, not with the seed's. That is why the LHS 1140 b tables carry `info`, $\|R\|$ and the composition drift in every row.

Two measured cases show that the seed choice is a real question of path and nothing more.

The helium scan needed no ladder. In the $K_{zz} = 10^9$ composition scan, every case restarted from the converged *diffusion-off* solution of the same composition, or from the neighboring composition already solved with diffusion on, and every case reached `info = 0` in its first *pass-set* and then stopped moving (LHS1140b/kzz_decision.md, section 6). The recorded reason is physical: a diffusion-off solution is well mixed near the base, which is where the adopted K_{zz} also mixes, so the step the line search is asked to take is small. The same held at 10^8 and 10^{10} (*ibid.*, section 6.1).

The eddy coefficient did need one. Restarting a case at $K_{zz} = 10^8$ or above directly from the $K_{zz} = 0$ control asks the solver to absorb the whole helium column in one step: the line search stalls and the composition outer loop stops on its solver-failure guard after a single pass (*ibid.*, section 3). Rebuilding one decade at a time (`ladder_kzz.sh` restarts each case from the case one decade below it) fixes the seeding. It does not fix everything: on this wind JFNK settles at $\|R\|$ between 2.7×10^{-3} and 4.7×10^{-3} rather than the 1.0×10^{-3} the control reaches, so the upper rungs state a looser `Resid tol` and quote the achieved $\|R\|$ in the table. A target of 2×10^{-2} was tried and rejected, because there the solver stops responding to the eddy term altogether.

Same code, same operation, opposite verdicts on whether a ladder is required, which is the concrete demonstration that seeding is about the step the nonlinear solver is asked to take, and that no property of the converged answer follows from it.

6. Evidence that the restart is faithful

6.1 The element projection conserves mass

Every write of an element total back into a species vector goes through `project_elements` (`binary_element_diffusion.f90`), and what it has to preserve is the mass the species handed in carried. Until 2026-09-10 it took the mixture mass as $m_{\text{sum}} = m_1 n_{\text{H}} + m_{\text{He}} n_{\text{He}}$ with m_1 the *reservoir* mass per hydrogen nucleus, which carries the trace metals at `melem_ab` while the cell carries them at whatever the trace transport left, and the difference was written into hydrogen at every call. MEASURED at one cell of the certification column, with metal/H 0.024401 against the reservoir's 0.028227: $\sum_i m_i f_i$ went $1.0000000 \rightarrow 1.0000636$ after one step, helium's gain exceeding the hydrogen and metal loss by 6.4×10^{-5} , which is mass created. A second leak of the same kind had the trace-metal transport writing metal densities with no counter-flux in the hydrogen they diffuse through (3.0×10^{-3} on the suite column with the metals frozen).

`mixture_mass_split` now reads the mass the cell's species actually carry, from `bsp_mass` and `melem_A` with no species weight restated, and `project_elements` scales component 1 as a whole, every helium-free species included, with the metals written first and the hydrogen group closing the mixture; one projection after the metal loop puts the metals' mass change back on hydrogen. Metals off is bitwise by construction ($m_1 = m_{\text{H}} = 1$ exactly). MEASURED: the certification column's closure after the relaxation went $3.37 \times 10^{-3} \rightarrow 2.3 \times 10^{-15}$, and the handed-back atomic candidate $6.44 \times 10^{-3} \rightarrow 1.94 \times 10^{-9}$, which is the file's own precision. Reported and not closed: the metal transport's counter-flux is closed on hydrogen but not on momentum or internal energy, which is a trace approximation.

6.2 The state survives the file

The writer-to-loader round trip returns the state. MEASURED on the 12000-step `mol_base_handoff` state with `EXHALE_DUMP_IC=1`, which writes the state as loaded: r , v and p come back bitwise, every species column and every inventory (H, He, O and C nuclei, charge, He/H, the H_2 ratio) to 4.2×10^{-16} , T to 5.6×10^{-16} , and a second trip to 5.0×10^{-16} with no drift. The ρ column comes back at 4.9×10^{-12} , and that is the writer's own mass closure and not the loader's: the loader never reads that column, it rebuilds the mass from the species, and perturbing the column by 10^{-15} changes nothing bitwise (the same state at 40 steps closes to 4.1×10^{-14}).

The stationary residual of a reloaded state was a separate matter and was closed later. Before the element projection was fixed to conserve mass (§6.1), the writer and the reload of one atomic candidate stood at $\|R\| = 3.2 \times 10^{-2}$ and 0.4385 on *different rows*; after it, at 1.112 and 1.1123 on the *same row*. The state is the same object on both sides of the file.

What the restart does *not* fix, and where the movement comes from. MEASURED, and worth separating from the loader: He/H departs from the input by 1.0×10^{-11} and 8.9×10^{-12} after 12000 marching steps with carrier transport off and on, but by 3.9 to 7.9 percent on a coupled-carrier density-unknown JFNK state and by 41 to 43 percent on an $\ln n$ state. The carrier column is written after the element projection precisely so that the Newton's carrier is not rescaled, the sweep reads the nuclei totals from the state, and with helium diffusion off no row re-imposes He/H. That is the

element constraint of the stationary system seen from the restart side; it is the producing path that moved the ratio, not the loader.

The diffused profile survives a round trip. Measured against the stored `examples/14_diffusion` output under `EXHALE_DUMP_IC=1`, which writes the state as loaded, before the first ionization-equilibrium solve: the loaded He/H profile reproduces the file's to 3.3×10^{-16} relative, worst cell, over all 504 cells (`docs/Update_EXHALE_stage1.pdf` §67). That is round-off, i.e. the `He_diffusion` branch of section 3. leaves the column it is meant to leave alone.

The defect it replaced. The same file through the pre-change binary comes back *flattened* to the input ratio: at $r = 1.05 R_p$ the file holds 1.616×10^{-2} and the load produced 8.333×10^{-2} , a factor 5.2 (*ibid.*). The old rule rescaled every cell to the global He/H whenever the file's composition differed from it, on the ground that the input file is the authority: correct with diffusion off, and destructive of the answer with diffusion on. Both numbers are quoted from that record and were not re-measured for this memo.

And the composition authority itself was a measured need. A He/H = 10 restart seeded from a He/H = 1 solution was measured to converge back to He/H = 1 everywhere before the redistribution existed (the measurement is recorded at the head of that block in `load_IC.f90`); the donor's composition ran while the setup report echoed the input one.

The diffusion-off path is unmoved. make check reported byte-identical output on every case of the matrix as it stood on 2026-08-25, which is the evidence that confining the rescale to the base cells changed nothing outside the `He_diffusion` branch (`docs/Update_EXHALE_stage1.pdf` §67, quoted; the matrix has since grown from five cases to sixteen, and was not re-run for this memo).

Part 2: the base handoff (scalars and profiles)

7. The scalar handoff

`read_base_inp` reads an optional `base.inp` from the run directory (`src/modules/files_IO/input_read.f90`: 1203–1388). It recognizes six named keys and one family of element keys, and every key belongs to one of the categories the routine's own header lists, because the category is what says what the value is allowed to do to the wind:

key	unit	category	what it sets
T_base	K	EOS boundary	T ₀ , the base temperature
r_base	R_J	EOS boundary	R ₀ , the base radius
p_base	bar	EOS boundary	p_base_bar, the level all of the above refer to (default 10 ⁻⁶)
q_H2_base	ratio	EOS boundary	q_h2_base, the H ₂ volume mixing ratio that sets the base particle count through comp_ntot_bc, in place of the chemical-equilibrium fit. An EOS anchor, not a composition pin: nothing holds H ₂ there and no H ₂ profile is seeded from it
HeH_base	ratio	elemental reservoir	HeH; a positive value also sets thereis_He
<El>_H_base	ratio	elemental reservoir	X _{El} , the El/H nuclei ratio, through set_element_abundance, for any of the ten elements of the species table (e.g. O_H_base 4.90e-4). Overrides metals.inp for that element, and switches the metal system on if it is the only nonzero abundance
Kzz_base	cm ² s ⁻¹	boundary constraint	he_kzz, the eddy coefficient the element-diffusion operator imposes at the base; inert with He_diffusion off

Comment lines and unknown keys are ignored, with one comment parsed: # solution_id <hash>, which is what the pair check of section 8. reads. The call sits in input_read after run_lower_atm_prestep and *before* the derived constants, so the overrides take effect on everything computed from them, and an absent file is a no-op (input_read.f90:858-882). Two consistency messages are printed and not enforced: that q_H2_base and HeH_base must come from the same lower-atmosphere solution (input_read.f90:1325-1336), and that a p_base away from 1 μ bar means the base composition and the equilibrium H₂ fit both refer to that other level (input_read.f90:1337-1342). Those two stay messages deliberately: an internally inconsistent scalar handoff is the user's to judge. What is *not* left to a message is the pair of files, see section 8..

Kzz_base deserves a note, because the scalar it writes has changed underneath it. he_kzz now defaults to 0, not 10⁹: an eddy term is a property of the atmosphere being modeled, so it is stated rather than inherited (src/modules/init/parameters.f90:146-156). And he_kzz is no longer what the diffusion operator reads. Every site in binary_element_diffusion reads the array kzz_cell instead (parameters.f90:157-164; the three sites are binary_element_diffusion.f90:679, 1380, 1597), filled once the radial grid exists by eddy_diffusion_on_grid:

from the interpolated Kzz column of a profile when there is one, and from `he_kzz` in every cell otherwise (`src/modules/files_IO/lower_atmosphere_profile.f90:465-529`). The uniform case is exactly the old scalar (for a uniform array the face average $\frac{1}{2}(a+a)$ is a in IEEE double), so a run with no profile sees the number it always saw.

A real file, the one the `mol_base_handoff` regression case pins (`backup/regression/mol_base_handoff/base.inp`, comments elided):

```
T_base      1140.0
r_base      0.49
HeH_base    0.0793
Kzz_base    1.000e+09
q_H2_base   0.75
p_base      1.000e-05
```

Four of the seven default regression cases carry such a file: `mol_base_handoff`, `mol_metals`, `mol_lyman_werner` and `mol_diffusion` (`backup/regression/run_check.sh:57`), so the scalar contract is pinned by the golden comparison and cannot be changed silently. Those four goldens were re-snapshotted on 2026-08-27, but for a reason that has nothing to do with the handoff: a molecular-network audit found the three-body term using a mass density where a number density belongs (`docs/Update_EXHALE_stage1.pdf §82`). The scalar contract itself is unchanged.

8. The profile handoff

The second route is the file named by `Lower atmosphere profile`: in `input.inp` (`input_read.f90:447-455`; user manual §3.8). It is a table over an interval of pressure rather than a single level, and the reader is `src/modules/files_IO/lower_atmosphere_profile.f90`.

What the file is. A header of `# key value` lines and then one row per level, deep to shallow. The header states the fingerprint (`solution_id`, `source_code`, `source_version`, `mechanism`, `stellar_flux`, `notes`), the three pressures (`p_match_bar`, `p_top_bar`, `p_deep_bar`), the closure iteration (`iteration`, `trial_flux_H`, `trial_flux_He`, `reached_steady_state`) and the column names (`lower_atmosphere_profile.f90:185-198`). Nine columns are required (`p`, `r`, `T`, `n_tot`, `rho`, `Kzz`, `q_H2`, `q_H`, `X_He`), and the element columns `X_<El>` are found by name, so the producers may order them as they like (`lower_atmosphere_profile.f90:328-340`).

What is refused. The reader refuses rather than warns on a non-positive or non-monotonic pressure, a table whose top does not reach above the matching level (`p_top_bar` $\geq$ `p_match_bar`, i.e. no overlap at all), a matching level outside the pressures the table covers, and a missing `solution_id` (`lower_atmosphere_profile.f90:344-382`). Interpolation is linear in $\log p$ and is never extrapolated; a matching level that falls on a node returns that node bit for bit, which is what lets a single-level profile reproduce the scalar handoff exactly (`lower_atmosphere_profile.f90:396-434`).

What EXHALE takes from it. Exactly five kinds of quantity, all through the same doors the scalar keys use (apply_lower_atmosphere_profile, input_read.f90:1392-1460): T_0 and R_0 from the T and r columns at the match, $p_{\text{base_bar}}$ from p_match_bar, $q_{\text{H}_2\text{_base}}$ from the q_H2 column, HeH from X_He, and each metal abundance from its X_<El> column through set_element_abundance, the same routine <El>_H_base goes through, which is why an element stated by either route is marked melem_from_handoff and picked up by the restart rescale of section 3.5. K_{zz} is taken as a profile, interpolated onto the radial grid into kzz_cell. The n_tot and rho columns are required of the file but are *not* imposed: the base density is n_0 of input.inp, and the base particle count and mass follow from T_0 , He/H, q_{H_2} and the elemental ratios exactly as on the scalar path.

The two files together are enforced, not printed. With a profile in use, every base.inp key in the three physics categories of section 7. (EOS boundary, elemental reservoir, boundary constraint) is refused with error stop, because the profile already states that quantity at the matching level and a scalar accepted beside it would be a second source of the same number (refuse_scalar_key, input_read.f90:1371-1386). What a base.inp beside a profile may still carry is provenance, and there the pair is checked: a missing # solution_id line, or one that differs from the profile's, stops the run (input_read.f90:1344-1367). He_Kzz: in input.inp is a different file with a different contract and is not refused, but it is inert beside a profile and the run says so (input_read.f90:1449-1458).

What pins it. The lower_profile regression case, one of the seven defaults (backup/regression/lower_profile/): HD 209458 b with He_diffusion: True, the base state and q_{H_2} taken from the table at $p_{\text{match}} = 10^{-6}$ bar, the elemental reservoirs carried by the profile with no_metals.inp in the directory (so the profile is the only source of the metal abundances), a non-constant Kzz column interpolated onto the grid in place of the inert scalar, and a provenance-only base.inp whose solution_id matches, which pins the accepting branch of the pair check. The worked example is examples/17_lower_profile/.

9. Scalar file against profile file

This is the comparison the choice was made on, brought up to what each route is in the code today.

	scalar base.inp	profile file
what is transmitted	six named numbers plus one El/H per element, all describing one layer	columns on the lower model's own pressure grid over an interval that reaches above the match
relation between the models	a handoff: the lower model's state at one level becomes the upper model's boundary	a matching over an interval both models cover, refused outright when the file does not reach above the match
flux continuity	not checkable. One layer's values carry no gradient, so nothing in the file constrains an elemental flux, and HeH_base stays a prescribed input while "the wind returns H:He" is an assumption dressed as a result	checkable, and measured. The elemental face flux $4\pi r^2(\rho X v + J)$ is reduced to a median and a relative spread over two radial windows and written to EXHALE_resolved.out whenever a profile is in use (binary_element_diffusion.f90:1784-1872); the criterion is the same radial invariance as T8 (docs/Update_EXHALE_stage1.pdf §69)
eddy coefficient	one constant, Kzz_base $\rightarrow$ he_kzz	the Kzz column, interpolated onto every cell as kzz_cell
molecular composition	one scalar, q_H2_base	q_H2 and q_H at their own levels, plus the diagnostic molecules; the ions of the EXHALE network are still not transmitted (section 10.3)
what it costs to produce	a keyword scan, unknown keys ignored	a schema with units and a grid convention, interpolation onto the EXHALE grid, a fingerprint and a stated failure policy, all written (src/utils/lower_profile_schema.py and the two adapters beside it)
failure mode	a silent mismatch: the two consistency checks are messages, not refusals (input_read.f90:1325-1342), so a handoff assembled from two different lower-atmosphere solutions runs without complaint, unless a profile is present, in which case the pair is checked by fingerprint	a stated one: a missing or unmatched solution_id, a scalar physics key beside the profile, a table that does not overlap the match, each an error stop with the reason printed

Three rows carried the decision.

Flux continuity is the whole point of Phase E. A steady lower-atmosphere chemistry solution cannot determine the hydrogen supply independently of its upper-boundary flux. Without a flux-continuity condition at a matching surface, HeH_base remains prescribed and the composition is an input wearing the clothes of an output. A scalar file cannot express the condition, because a single level has no gradient and carries no flux. This is not a convenience argument; it

is the difference between a coupling and an assumption.

A constant K_{zz} is the wrong shape. The judgment is already on record: “a *constant* K_{zz} is the wrong shape of assumption for this problem regardless of its magnitude: the molecular coefficient it competes against rises by nearly five decades between the base and $1.2R_p$, so no single constant can both mix the base and follow the molecular coefficient outward, while Lindzen’s own result is that the eddy coefficient should *turn over* rather than keep growing” (docs/eddy_diffusion_kzz.tex, summary). K_{zz_base} can transmit only a constant. The homopause (the one consequential thing K_{zz} does) is set where $K_{zz} = D$, so it is a property of the two profiles and not of either number alone.

The molecular scalar has the same defect one level down. $q_{H_2_base}$ is a single H_2 mixing ratio standing in for a composition that the molecular network resolves into H_2 , H_2^+ , H_3^+ and HeH^+ . It is enough to select a branch of the base particle count; it is not enough to hand over a photochemical solution. This is the one row of the three the profile route did not close: it transmits q_{H_2} and q_H at their own levels, but none of the three molecular ions (section 10.3).

10. The decision, and what was built on it

10.1 Decided 2026-08-26 (user decision)

1. **Phase E goes to a profile file.** The interface decision the plan left open (“whether this stays a scalar file or becomes a profile file is an interface decision to confirm with the user before implementation”) is settled in favor of profiles, on the flux-continuity row of section 9. above all. The plan itself now records the phase as done (milestones E1–E5, 2026-08-27) with its three acceptance criteria and the numbers behind them.
2. **The scalar base.inp is kept**, as the single-layer route rather than a deprecated one. Three reasons, in order: four regression cases pin it and their goldens are the evidence that the molecular base is unmoved; a single-level handoff remains the right description when the lower model genuinely produces one (the analytic column of Tier 1); and the code path is small and already validated.
3. **The profile route is opt-in through a new key** (`Lower atmosphere profile:`), and its absence leaves the previous behavior exactly as it was. This is the standing house rule: a new physics option defaults to off so old goldens still reproduce.

10.2 What was implemented, 2026-08-27

All three hold, with one qualification on the third. The route is opt-in through `Lower atmosphere profile:` and an absent key is a no-op, so the diffusion-off and scalar-handoff paths are untouched by any of it. The qualification is that the four molecular goldens are *not* the ones they were at the time of the decision: they were re-snapshotted at the end of the same day for the three-body density fix of docs/Update_EXHALE_stage1.pdf §82, which is a different change with a different reason. The Phase E work did not move them.

What the decision left open, and where it now lives:

piece	where it is
the schema	md/input_schema.md §2d; user manual §3.8
the reader	src/modules/files_IO/lower_atmosphere_profile.f90
the run key	Lower atmosphere profile: (input_read.f90:447-455)
the producers	src/utls/lower_profile_schema.py (format, elemental accounting, fingerprint) with photochem_to_lower_profile.py and vulcan_to_lower_profile.py behind one command line
the closure loop	src/utls/element_flux_closure.py, driven from the fluxes EXHALE measures and writes to EXHALE_resolved.out
the regression	backup/regression/lower_profile/, one of the seven defaults
the example	examples/17_lower_profile/

10.3 What the profile file carries, against what the plan asked for

The plan's list (*ibid.*, Phase E part 1) against the schema, field by field. "read by" names the code that consumes it; a field the file carries and nothing consumes is stated as such rather than counted as met.

asked for	field	read by
matching pressure and radius	header <code>p_match_bar</code> ; columns <code>p</code> , <code>r</code>	<code>p_base_bar</code> directly; <code>R0</code> from <code>r</code> at the match (<code>input_read.f90:1418-1424</code>)
temperature and total density	columns <code>T</code> , <code>n_tot</code> , <code>rho</code>	<code>T0</code> from <code>T</code> . <code>n_tot</code> and <code>rho</code> are required of the file but not imposed: the base density stays <code>n0</code> of <code>input.inp</code> and the base particle count and mass are built from <code>T0</code> , <code>He/H</code> , <code>q_{H2}</code> and the elemental ratios, as on the scalar path
elemental H and He abundances	columns <code>X_He</code> , <code>X_<El></code>	<code>HeH</code> and, through <code>set_element_abundance</code> , every <code>melem_ab</code> (<code>input_read.f90:1430-1443</code>)
the upward elemental fluxes	header <code>trial_flux_H</code> , <code>trial_flux_He</code> ; columns <code>F_H</code> , <code>F_He</code>	the header pair by <code>element_flux_closure.py</code> , through <code>EXHALE_resolved.out</code> . The <code>F_H/F_He</code> <i>columns</i> are written by the adapters and read by no Fortran routine: <code>EXHALE</code> measures its own elemental fluxes instead and reports them, which is the half of the loop it owns
molecular/atomic H partition	columns <code>q_H2</code> , <code>q_H</code>	<code>q_H2</code> → <code>q_h2_base</code> , the base particle count. <code>q_H</code> is required of the file and read by nothing
K_{zz} at the match	column <code>Kzz</code>	not the match alone: the whole column, onto <code>kzz_cell</code> (<code>eddy_diffusion_on_grid</code>)
selected molecular abundances the EXHALE network needs	<code>q_H2</code> , plus the diagnostic set (<code>q_H2O</code> , <code>q_CO</code> , <code>q_CH4</code> , ...)	<i>not met</i> . The <code>EXHALE</code> network resolves <code>H2</code> , <code>H2⁺</code> , <code>H3⁺</code> and <code>HeH⁺</code> ; the file carries none of the three ions and no molecular profile is seeded from it. The diagnostic columns are metadata (nothing consumes them), and the schema says so
source fingerprint	header <code>solution_id</code> , <code>source_code</code> , <code>source_version</code> , <code>mechanism</code> , <code>stellar_flux</code> , <code>iteration</code> , <code>notes</code>	the reader, and <code>lap_report_provenance</code> into <code>EXHALE_resolved.out</code>

The two further requirements the plan states are both met:

- **the same-solution consistency of the handoff pair is enforced**, not printed. With a profile in use a `base.inp` without a `# solution_id` line, or with one that differs from the profile's, stops the run, and a scalar physics key beside the profile stops it earlier still (`input_read.f90:1344-1367`, `1371-1386`). Note the scope: this is enforcement of the *pair*. A `base.inp` on its own is still checked only by the two printed messages of section 7.;
- **the transit tool reads the resolved configuration**. `read_input_parameters` prefers `EXHALE_resolved.out` next to `input.inp` and takes the planet radius, mass, equilibrium temperature, orbital distance, stellar mass

and He/H from it, refusing to fall back silently if the file is present but unreadable (`exhale_transit_lib.py:336-364`). `EXHALE_resolved.out` is written after both handoff readers have run (`src/modules/files_IO/write_setup_report.f90:428-543`), so a handoff that moves the base radius now moves the transit geometry with it, Phase B of the plan.

10.4 What is still open

- The molecular ions of the network are not part of the handoff, as above. A photochemical $\text{H}_2/\text{H}_2^+/\text{H}_3^+/\text{HeH}^+$ solution at the base is still something EXHALE re-derives rather than inherits.
- The overlap window the design defines the closure on was measured on LHS 1140 b and rejected there: it is $\sim 0.003 R_p$ thick and the flux across it spreads by a factor of 13–35, so the driver falls back on the steady-flux window $r \geq r_{\text{esc}}$ and records that it did (`src/utls/element_flux_closure.py`, header). The fallback is reported rather than silent, but the window the design was written around is not the window the number comes from.
- `q_H2_base` and the profile's `q_H2` are the composition of the inflowing gas in both routes, no longer EOS anchors alone (§117 of `docs/Update_EXHALE_stage1.tex`, 2026-09-01). They set the base particle count *and* are imposed on the H_2 partition of the lower ghost species, the particle count then coming from that same species state. Before that change the two disagreed by 6.4% and the base ran 73 K above the T_0 the boundary condition claimed to impose.